



AWS ROUTE 53

★ COMPLETE NOTES ★

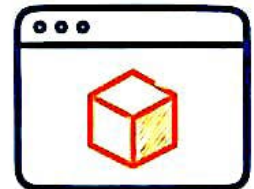
BEGINNER → ADVANCED



USER



ROUTE 53
(DNS)



AWS
APPLICATION



DNS



ROUTING



HEALTH
CHECKS



DOMAINS

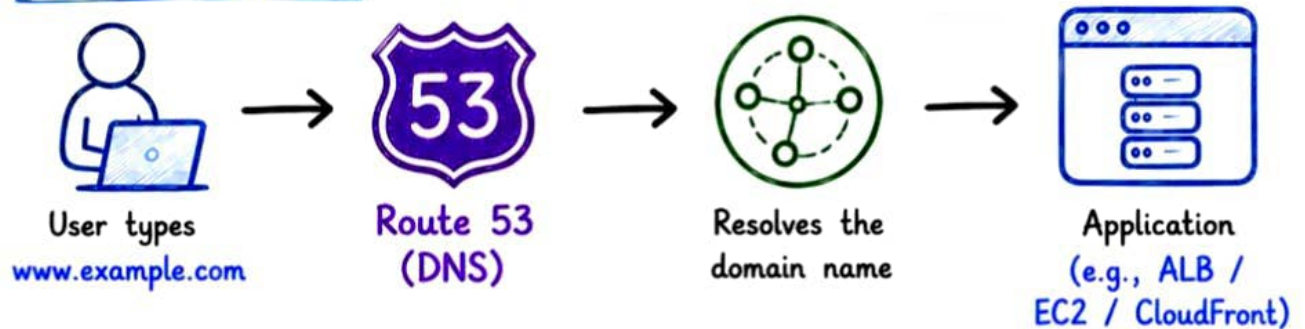


DNS + ROUTING + HEALTH CHECKS —
MADE EASY

2 WHAT IS AWS ROUTE 53?

Amazon Route 53 is AWS's highly available and scalable **DNS** service.

HOW IT WORKS?



CORE FUNCTIONS



ROUTE 53 CAN ROUTE TRAFFIC TO



**MEMORY LINE:**
Route 53 is the **PHONEBOOK** + **TRAFFIC DIRECTOR** of your applications on the internet.



3 DNS RECORDS IN ROUTE 53

DNS records tell Route 53 how to **route traffic** for a domain or subdomain.

RECORD TYPE	PURPOSE	EXAMPLE
A	Maps domain or subdomain to an IPv4 address.	example.com ↓ 192.0.2.10
AAAA	Maps domain or subdomain to an IPv6 address.	example.com ↓ 2001:db8::1
CNAME	Maps one domain name to another domain name.	www.example.com ↓ example.com
MX	Specifies mail servers for the domain.	example.com ↓ 10 mail.example.com
TXT	Stores text information. Used for verification, SPF, DKIM, etc.	example.com ↓ "v=spf1 include:spf.protection.outlook.com ~all"
NS	Specifies the name servers for the domain.	example.com ↓ ns-123.awsdns-45.com ns-987.awsdns-65.net
ALIAS	Routes traffic to AWS resources. Can be used at the zone apex (example.com).	example.com ↓ ALB / CloudFront / S3 / API Gateway

WHY ALIAS?

- ✓ Works at zone apex
- ✓ No extra cost
- ✓ Automatically tracks IP changes
- ✓ Highly available

ALIAS EXAMPLE



CNAME vs ALIAS

CNAME ✗

- Cannot be used at zone apex
- Extra DNS lookup

ALIAS ✓

- Can be used at zone apex
- Direct routing to AWS resources



QUICK TIP

Use **ALIAS** to point your domain to AWS resources whenever possible.



MEMORY TRICK

A for Address, **AAAA** for Address (v6),
CNAME for Change Name,
MX for Mail eXchange!

4 ROUTE 53 ROUTING POLICIES

Routing policies help Route 53 decide **WHERE** to send the traffic.

CHOOSE THE **RIGHT** POLICY FOR YOUR USE CASE

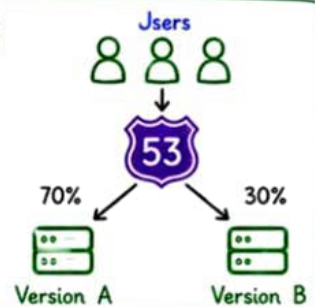
1 SIMPLE ROUTING

- Routes traffic to one resource.
- Use when you have a single resource.



2 WEIGHTED ROUTING

- Splits traffic across multiple resources based on weights.
- Useful for A/B testing, blue/green deployments.



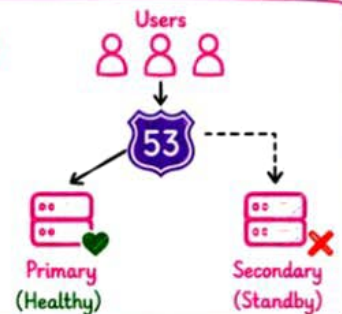
3 LATENCY-BASED ROUTING

- Routes users to the Region with the lowest latency.
- Improves performance for global applications.



4 FAILOVER ROUTING

- Routes to primary resource.
- If primary fails health check, traffic is routed to secondary.



5 GEOLOCATION ROUTING

- Routes traffic based on the geographic location of the users.
- Useful when content or applications differ by location.



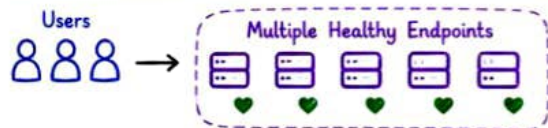
6 GEOPROXIMITY ROUTING

- Routes based on the geographic location of users & resources.
- Uses bias to shift more traffic to preferred locations.



7 MULTIVALUE ANSWER ROUTING

- Returns multiple healthy records (up to 8) in response to DNS query.
- Works with health checks.
- Not a substitute for load balancers.



PRO TIP
Pick the routing policy that matches your application goal and architecture.

WHEN TO USE WHAT?

- Simple → Single resource
- Weighted → Traffic split / Testing
- Latency → Global performance
- Failover → DR / High availability
- Geolocation → Content by country
- Geoproximity → Advanced geo routing
- Multivalue → Multiple healthy records

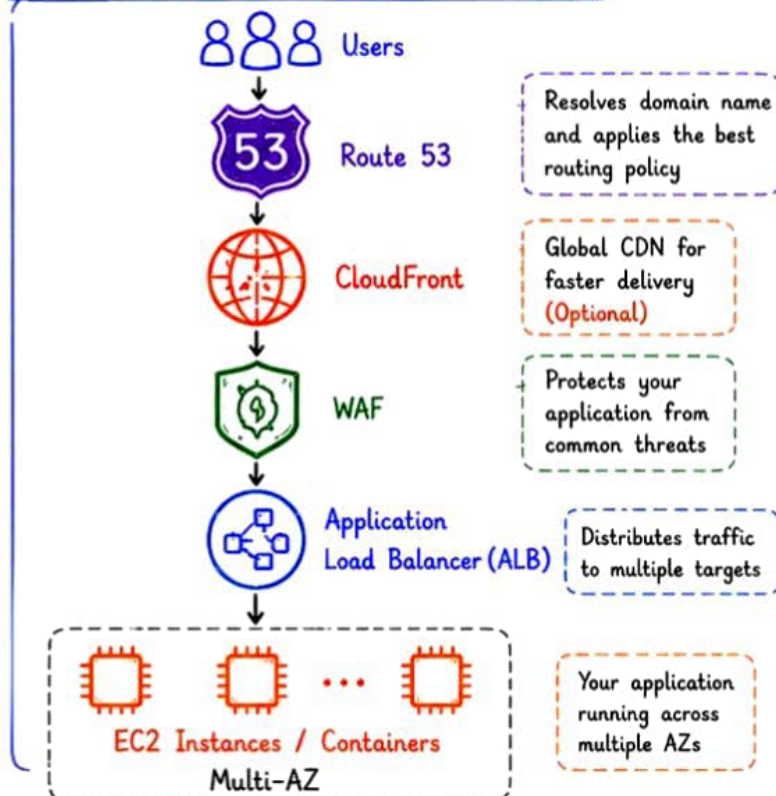
REMEMBER

The right routing policy = **Better performance + High availability!**

5 ROUTE 53 IN REAL AWS

Route 53 is the entry point for your application.
It works behind the scenes to route users to the **right place!**

COMMON PRODUCTION ARCHITECTURE



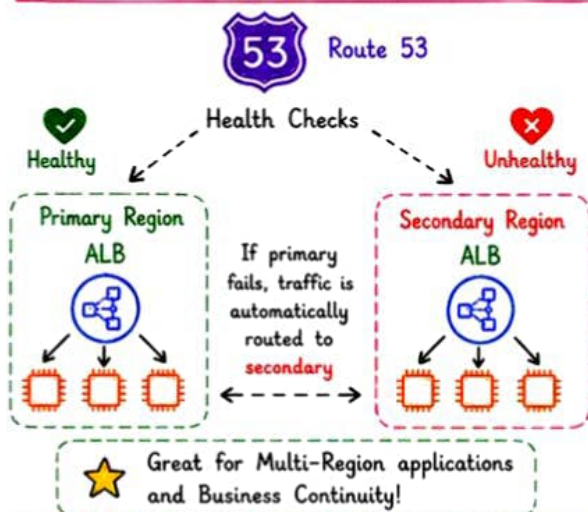
WHY THIS ARCHITECTURE?

- ✓ High Availability
- ✓ Scalability
- ✓ Security
- ✓ Global Performance
- ✓ Fault Tolerance

USE CASES

- Web Applications
- E-commerce Websites
- APIs and Microservices
- Media & Streaming Apps
- Global Applications

DISASTER RECOVERY WITH ROUTE 53



ROUTE 53 + AWS SERVICES



ELB (ALB / NLB)
Route traffic to your Load Balancers



CloudFront
Global content delivery



API Gateway
Route to your APIs



S3 Website
Host static websites



EC2 / Auto Scaling
Route to EC2 instances



Other AWS Services
PrivateLink, VPC Endpoints, etc.



KEY TAKEAWAY

Route 53 does **DNS + Routing** decisions, while other AWS services handle the traffic!



6 ROUTE 53 CHEAT SHEET

QUICK RECAP



DNS

Converts domain names to the right destination.



HEALTH CHECK

Monitors the health of your endpoints continuously.



ROUTING POLICY

Decides where and how traffic should be routed.



HOSTED ZONE

Container for DNS records for your domain.



ALIAS RECORD

Route traffic to AWS resources at zone apex.

INTERVIEW QUESTIONS

1. What is Amazon Route 53?
2. What are the components of Route 53?
3. What is the difference between A record and CNAME record?
4. What is an Alias record and when would you use it?
5. What are the different routing policies in Route 53?
6. How does Route 53 health check work?
7. What is the difference between Public and Private Hosted Zones?
8. How would you design a multi-region failover using Route 53?

TYPICAL FLOW



USER

Enters domain in browser



ROUTE 53

Resolves the domain using DNS



ROUTING POLICY

Applies the right routing logic



HEALTH CHECK

Chooses a healthy endpoint



DESTINATION

Traffic reaches your application

BEST PRACTICES

- ✓ Use appropriate routing policy for your use case.
- ✓ Always enable Health Checks for critical endpoints.
- ✓ Use Alias records to route to AWS resources.
- ✓ Separate environments using subdomains.
- ✓ Monitor traffic and health regularly.



MEMORY TIP

Think of Route 53 as the **TRAFFIC DIRECTOR** of the internet.

Right user → Right place → Right time



★ SAVE THIS SLIDE FOR YOUR AWS INTERVIEW PREPARATION! ★

10 BEST PRACTICES FOR ALB

Follow these best practices to build **secure**, **scalable** and **cost-effective** ALB.



① Use HTTPS listeners and redirect HTTP to HTTPS.



② Enable cross-zone load balancing for even traffic distribution.



③ Use target group health checks with proper thresholds.



④ Use sticky sessions only when your application requires it.



⑤ Keep your ALB and target groups in the same VPC.



⑥ Use WAF to protect against common web exploits.



⑦ Monitor with CloudWatch and set alarms.

COMMON MISTAKES

- ✗ Using HTTP instead of HTTPS.
- ✗ Not configuring health checks properly.
- ✗ Registering unhealthy instances.
- ✗ Ignoring idle timeout settings.
- ✗ Not enabling access logs.
- ✗ Placing ALB in a public subnet without security.



IMPORTANT CLOUDWATCH METRICS



- RequestCount
- TargetResponseTime
- HTTPCode_ELB_5XX_Count
- HTTPCode_Target_5XX_Count
- HealthyHostCount
- UnHealthyHostCount
- ConsumedLCUs

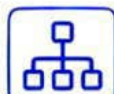
DESIGN CHECKLIST



Users



Internet



ALB



Target Group



Targets



Monitoring

✓ Identify users & traffic needs

✓ Ensure DNS is configured

✓ Create ALB with secure listeners

✓ Configure TG with proper health checks

✓ Register healthy targets

✓ Enable monitoring & logs

KNOWLEDGE RECAP

ALB

→ Intelligent routing

→ Multiple targets

→ High availability

→ Better performance

→ Happy users!



MEMORY TRICK:

A Smart Balance = High Availability + Speed + Security

@NeerajSingh-DevOps